



# A multimodal deep learning framework for stock direction prediction with granger-based predictive explainability

Mohd Raagib Shakeel<sup>1</sup> · Satyam Yadav<sup>2</sup> · Piyush<sup>2</sup> · Musheer Ahmad<sup>3</sup>

Received: 2 December 2025 / Accepted: 7 April 2026

© The Author(s), under exclusive licence to Academy of Economics and Finance 2026

## Abstract

Predicting stock price movements remains challenging due to the noisy, nonlinear, and multimodal nature of financial markets. While existing studies combine technical indicators with news sentiment, they often lack transparency and fail to uncover temporal predictive relationships across modalities. This paper proposes a predictively explainable multimodal forecasting framework that integrates structured price-based features and unstructured financial news using FinBERT embeddings within a recurrent neural network. A dual explainability scheme is employed: Integrated Gradients for feature attribution and Neural Granger causality to reveal temporally ordered predictive relationships. Using 17 years of Nifty 50 data (2003–2019) aligned with 73,500 financial news headlines, the model is evaluated under a strict rolling-origin time-series cross-validation protocol to prevent data leakage. Results demonstrate superior predictive accuracy, significantly outperforming a comprehensive set of statistical, machine-learning, and financial benchmark models, with significance validated via the White Reality Check. An ablation study confirms news embeddings as the dominant predictive modality. Granger-based analysis further reveals that specific semantic dimensions systematically predict future returns and volatility. A transaction-cost-aware trading simulation confirms the model's economic relevance, yielding economically meaningful risk-adjusted performance. The study presents a scalable, interpretable, and robust framework for financial forecasting, offering insights for both academic research and real-world decision-support systems.

**Keywords** Multimodal financial prediction · Financial news sentiment · Explainable Artificial Intelligence (XAI) · Time series prediction · FinBERT · Granger causality

Extended author information available on the last page of the article

Published online: 25 April 2026

Springer

## 1 Introduction

The predictability of stock prices lies at the core of financial economics, directly challenging the Efficient Market Hypothesis which posits that prices fully reflect all available information. Persistent forecasting efforts are motivated by observable market inefficiencies, behavioral biases, and gradual information diffusion. Classical technical analysis relies on indicators such as moving averages, RSI, and MACD (Lo et al. 2000), yet these backward-looking signals often fail to capture exogenous shocks and shifts in investor sentiment. Consequently, research has moved toward multimodal forecasting frameworks that combine market data with financial news (Liu 2018; Schumaker and Chen 2009). Although such hybrid deep learning models improve predictive accuracy, their increasing complexity—particularly in LSTMs and transformers raises interpretability concerns in financial applications where transparency and accountability are essential. Due to the nonlinear structure of deep learning models, various Explainable AI (XAI) methods have been proposed. Attention mechanisms, SHAP, and LIME (Muhammad et al. 2024; Çalik et al., 2025; Marey et al. 2024) provide local feature attributions but do not capture temporally ordered predictive relationships or structured cross-modal dependencies. Even advanced architectures such as geometric hypergraph attention networks (Harit et al. 2024) struggle to distinguish predictive influence from spurious association. This limitation constrains the adoption of AI-based forecasting models in financial decision-making.

More recent work has attempted to incorporate causal structure into financial prediction models. CausalStock integrates temporal inter-stock relationships to enhance predictive interpretability (Li et al. 2024), while MSGCA proposes gated cross-attention for multimodal fusion (Zong and Zhou 2024). However, these frameworks either exclude textual sentiment or lack formal econometric validation of predictive relationships. Established approaches such as Granger causality and Fast Causal Inference provide principled tools for distinguishing predictive precedence from structural dependence (Granger 1969; Tank et al. 2018; Feng 2020), yet their integration into multimodal deep learning pipelines remains limited. Many recent models treat text and numerical inputs as loosely coupled streams without structured interdependency modeling (Zhang et al. 2023; Lee and Tan 2022), while local attribution tools such as SHAP and LIME (Ribeiro et al. 2016; Nguyen et al. 2023) fail to capture temporally ordered predictive relationships (Huang et al. 2024). Although time-series causal inference methods show promise (Runge et al. 2019; Kalteh et al. 2022), they remain underutilized in multimodal deep learning research.

In order to address these two significant issues, the current study suggests an interpretable deep learning model for binary (0,1) stock price movement prediction, which incorporates both numerical data and unstructured text data. Unlike previous models that prioritize predictive accuracy, undermining interpretability, the suggested framework combines stock price data along with technical indicators with financial news using FinBERT and recurrent neural networks (RNN), both exhibiting a dual-layered explainability framework. This framework uses integrated gradients for fine-grained, modality-aware attribution and applies Granger causality analysis to identify temporal dependencies. In the process, the model not only improves forecast accuracy but also provides actionable, economically interpretable insights into the temporally

ordered predictive relationships between news sentiment and stock price movements, thereby contributing to a deeper understanding of information processing in financial markets. The key contributions of this research are outlined below:

1. A unified input pipeline is proposed that combines the structured numerical data (stock price & technical indicators) and unstructured financial news using FinBERT and RNN. This is complemented with PCA-based dimensionality reduction to reduce redundancy and noise between modalities—enhancing efficiency and performance in long-sequence modeling.
2. A double-layered explainability layer is proposed in this study by integrating Integrated Gradients (IG) and Granger Causality Analysis. The IG enables fine-grained feature attribution across both time and modalities, which provides detailed insights into the relative importance of input features. Granger Causality Analysis is employed to identify the temporal dependencies among features that provide insights into the underlying temporal causal (predictive) dynamics. This unification resolves a primary weakness in current multimodal financial models; wherein high predictive accuracy has been achieved at the cost of transparency and predictive interpretability.
3. The Cross-domain Granger-causal (predictive) graphs are constructed and visualized, demonstrating how news sentiment Granger-causes movements in numerical data such as returns, volatility. This offers actionable transparency into the directional influence of unstructured textual signals on market behavior.
4. The Rolling-window analysis and market regime segmentation (bullish, bearish, neutral, and high-volatility) are implemented, whose results indicate that news features consistently dominate prediction influence, especially in longer temporal lags. This highlights the need to prioritize semantic memory from textual data in financial modeling.
5. The model performance is statistically validated using the White Reality Check which confirms that the proposed RNN-based model significantly outperforms LSTM and GRU baselines without overfitting or data-snooping bias that demonstrates model's strong generalizability.
6. This study offers one of the most comprehensive and longitudinal evaluations of financial multimodal learning for the Indian market, which is an underexplored but globally relevant market, by using 17 years (2003–2019) of Nifty 50 index data aligned with over 73,500 financial news headlines from Business Standard.

The following sections are outlined as follows: Sect. 2 gives a comprehensive overview of the related work. Section 3 outlines the suggested framework, explaining the conceptual model, the corresponding mathematical structure underlying the analytical strategy, and the data gathering and research methodology of the empirical study. Section 4 contains the findings and analyses using Recurrent Neural Networks (RNN), model verification by the White Reality Check, and explainability of model predictions through Explainable AI (XAI) through the feature attribution approach of predictive relations analysis through Granger causality tests. Section 5 provides the discussion and Sect. 6 provides the conclusion and possible future research directions.

## 2 Literature review

This study builds on three interconnected research streams: financial forecasting and market efficiency, multimodal deep learning, and explainable AI (XAI) with causal inference. Recent research recognizes that asset prices are shaped by both numerical and textual information. This section reviews these strands and highlights their limited integration within causal financial modeling.

### 2.1 Foundational work in financial forecasting and causality

The effort to predict asset returns has a long history in financial econometrics. (Lo et al. 2000) laid the statistical groundwork for technical analysis, while (Granger 1969) introduced the concept of causality to test predictive relationships and information flow in time series data, core issues in market efficiency. Early studies that incorporated news relied on basic keyword counts or sentiment scores (Tetlock 2007), showing that media content influences market movements and that news sentiment is a priced risk factor. However, these approaches relied on relatively simple text processing and linear models, limiting their ability to capture contextual semantics that modern language models such as FinBERT can encode (Araci 2019). Our study revisits this problem using contemporary representation learning and predictive causal analysis.

### 2.2 Multimodal prediction: Numerical and textual inputs

Building on earlier evidence that news affects financial markets, recent deep learning studies have explored more complex relationships. For example, (Liu 2018) proposed an attention-based LSTM for index forecasting, while AZFinText (Schumaker and Chen 2009) demonstrated the predictive value of financial news. More recent studies combine FinBERT embeddings with deep architectures (Gu et al. 2024; Karadaş et al. 2025; Khubiev and Semenov, 2025), while other deep learning approaches such as dense residual networks have also been applied to financial prediction tasks including net asset value forecasting (Shakeel et al. 2023), often reporting improved predictive performance but limited economic interpretability. Explainable deep learning frameworks integrating SHAP and LIME have been introduced to identify influential features (Muhammad et al. 2024; Çalik et al. 2025; Marey et al. 2024), and broader deep learning applications continue to expand in financial modeling (Aminimehr et al. 2024). However, most of these approaches rely on local attribution methods and do not model structured cross-modal temporal dependencies or formally test predictive causal relationships.

### 2.3 Causal and structured explainability

Recent research has incorporated causal reasoning and explainable AI techniques into financial forecasting. For example, recent studies have integrated deep learning with econometric frameworks such as GRU and VAR models to analyze relationships between financial volatility indices and improve interpretability of

trading strategies (Shakeel et al. 2025). (Luciano et al. 2024) extended Shapley–Lorenz values to measure temporal influence, while (Harit et al. 2024) used hypergraph attention with SHAP to visualize dependencies in news embeddings. (Xu et al. 2024) constructed causal graphs from textual inputs, and (Feng 2020) explored nonlinear causal inference using PCA to address latent confounding. (Kleinberg et al. 2010) proposed a temporal logic-based causal framework linking stock prices and news through Granger causality, establishing a foundation for structured multimodal causal reasoning. While multimodal models improve predictive performance, XAI methods typically provide localized explanations without capturing global temporal structure, and causal graph approaches often remain confined to a single modality.

To overcome these limitations, our study develops a unified model that bridges this methodological divide. We combine FinBERT-based news embeddings with technical indicators within an RNN architecture. For explainability, we integrate a technique from deep learning (Integrated Gradients) with a cornerstone of econometric causal inference (Granger causality tests). This dual approach offers both local interpretability and global, econometrically-grounded causal insights of how structured and unstructured signals interact with each other to drive stock movement predictions.

### 3 Proposed framework

The proposed framework develops a predictively interpretable (Granger-based) multimodal model for binary stock price movement prediction by integrating structured numerical features with FinBERT-generated financial news embeddings (Araci 2019) within a recurrent neural network (RNN) architecture (Graves 2012). To enhance interpretability, a dual explainability mechanism is employed: Integrated Gradients quantifies feature-level attribution, while Granger causality analysis identifies temporally ordered predictive relationships among variables. Together, these components provide both local attribution and global causal (predictive) insight for financial decision-support. Figure 1 presents the schematic structure of the framework.

#### 3.1 Mathematical framework

We formalize the proposed framework through a structured mathematical model that defines the feature representation, fusion process, predictive architecture, and interpretability modules. This formalization ensures reproducibility and provides a clear basis for the empirical analysis.

##### 3.1.1 Step 1: Notation and data generating process

Let  $\{t\}_{t=1}^T$  denote trading days. For each day  $t$  we observe:

- Structured price features:  $x_t \in \mathbb{R}^p$  (e.g., log-return, volatility, MA\_5, MA\_10, RSI, etc.).

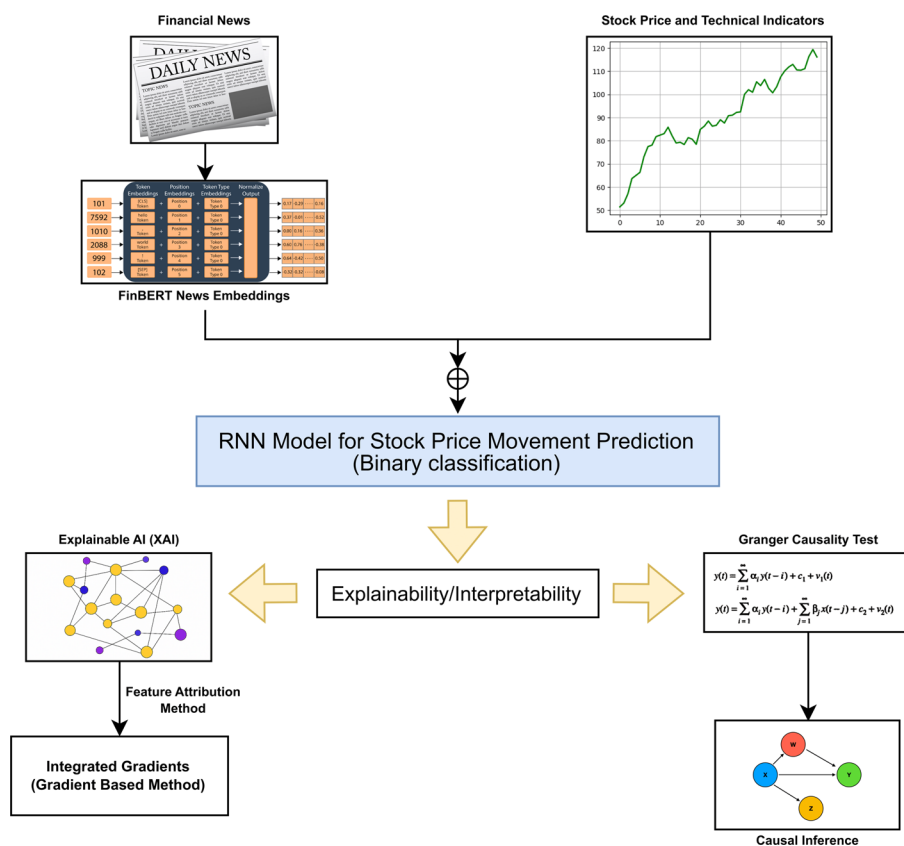


Fig. 1 Proposed Framework

- A set of textual items (news headlines)  $\mathcal{N}_t = \{n_{t,1}, \dots, n_{t,m_t}\}$  published on day  $t$ .
- Binary target  $y_t \in \{0,1\}$  indicating upward movement (1) or not (0) from close  $t-1$  to close  $t$ .

Assume there exists an underlying (unknown) stochastic process for the observable vector  $Z_t \triangleq (x_t, \mathcal{N}_t, y_t)$  generated by some stationary ergodic process under regimes discussed later.

Table: Notation.

- $p$  : dimension of numerical features.
- $d$  : raw dimension of single FinBERT embedding ( 768).
- $m_t$  : number of news items at time  $t$ .
- $k$  : look-back window length for sequence models.
- $r$  : PCA reduced dimension after multimodal projection.
- $f_\theta(\cdot)$  : parameterized predictive neural map.
- $\mathcal{L}(\cdot, \cdot)$  : loss function (binary cross-entropy).

### 3.1.2 Step 2: Representation of textual inputs (FinBERT) and aggregation

Each headline  $n_{t,j}$  is mapped by a pre-trained encoder (FinBERT) to an embedding as shown in Eq. 1:

$$e_{t,j} = \phi_{BERT}(n_{t,j}) \in \mathbb{R}^d \quad (1)$$

Item-level embeddings are aggregated into a single daily textual vector through a permutation-invariant aggregation operator  $A : \mathbb{R}^{m_t \times d} \rightarrow \mathbb{R}^d$  as shown in Eq. 2:

$$u_t = A\left(\{e_{t,j}\}_{j=1}^{m_t}\right) = \text{softmax}(\alpha_t) \odot \sum_{j=1}^{m_t} w_{t,j} e_{t,j} \quad (2)$$

where  $w_{t,j}$  is optional learned attention weight and  $\alpha_t$  a normalization. Special cases: simple average, volume-weighted average, or temporal attention.

### 3.1.3 Step 3: Multimodal feature construction and dimensionality reduction

Form the concatenated raw feature vector:  $v_t = \begin{bmatrix} x_t \\ u_t \end{bmatrix} \in \mathbb{R}^{p+d}$

To reduce dimensionality and control estimation error, apply PCA (empirical) to a training set  $\{v_t\}_{t \in \mathcal{T}_{\text{train}}}$ . Let  $P_r \in \mathbb{R}^{r \times (p+d)}$  denote the matrix of the first  $r$  principal components (columns are orthonormal) (Eq. 3).

$$z_t = P_r v_t \in \mathbb{R}^r \quad (3)$$

$z_t$  is the reduced multimodal representation used as input to dynamic models. Principal-component selection: choose  $r$  by explained variance > threshold (e.g., 90%) or crossvalidated predictive performance.

### 3.1.4 Step 4: Predictive model — recurrent state-space formulation

Let  $z_t \in \mathbb{R}^r$  be the reduced multimodal input at time  $t$  (as defined in Sect. 3). Introduce a hidden state vector  $s_t \in \mathbb{R}^h$  representing the model's latent information about past observations. The model dynamics follow a recurrent state-space system as shown in Eq. 4:

$$s_t = \Psi_{\theta}(s_{t-1}, z_t), s_0 = 0 \quad (4)$$

where  $\Psi_{\theta}$  is a nonlinear transformation parameterized by  $\theta$ , such as an RNN, GRU, or LSTM cell.

The observation (output) equation is defined as shown in Eq. 5:

$$\hat{p}_t = g_{\theta}(s_t) = \sigma(Vs_t + c) \quad (5)$$

where  $V \in \mathbb{R}^{1 \times h}$  and  $c \in \mathbb{R}$  are output-layer parameters, and  $\sigma(\cdot)$  is the logistic (sigmoid) activation mapping hidden states to probabilities  $\hat{p}_t = \Pr(y_t = 1 \mid \mathcal{F}_{t-1})$ . The parameter set  $\theta$  (including all recurrent and output weights) is estimated by minimizing the regularized empirical cross-entropy loss over the training set  $\mathcal{T}_{\text{train}}$  (Eq. 6).

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \frac{1}{|\mathcal{T}_{\text{train}}|} \sum_{t \in \mathcal{T}_{\text{train}}} \left[ -y_t \log \hat{p}_t - (1 - y_t) \log (1 - \hat{p}_t) \right] + \lambda \|\theta\|_2^2 \quad (6)$$

where  $\lambda > 0$  is the regularization coefficient controlling model complexity. Optimization is carried out using stochastic gradient-based algorithms (e.g., Adam) with early stopping to prevent overfitting.

### 3.1.5 Step 5: Identification and probabilistic assumptions

To permit meaningful inference and ensure model identifiability, the following assumptions are imposed:

- A1 (Stationarity and Ergodicity). The joint process  $\{(v_t, y_t)\}$  is strictly stationary and ergodic within each regime, or locally stationary under a rolling window approach.
- A2 (Weak Dependence/Mixing). The process is  $\alpha$ -mixing with coefficients decaying at a geometric rate, ensuring the applicability of LLN and CLT to gradient-based estimators.
- A3 (No contemporaneous unobserved confounders). Conditional on observed lags and covariates, no omitted variables contemporaneously affect both inputs and outputs.
- A4 (Model approximation). The recurrent neural network class is sufficiently expressive to approximate the true conditional probability function  $\Pr(y_t = 1 \mid \mathcal{F}_{t-1})$  arbitrarily well as model capacity increases.

These assumptions are standard for consistent estimation in nonlinear time series models and permit asymptotic justification for empirical risk minimization.

### 3.1.6 Step 6: Feature attribution: Integrated gradients (IG)

Let  $F_\theta(v_{t-k+1:t})$  denote the scalar logit output from the trained model given a sequence of multimodal inputs of length  $k$ . The Integrated Gradients (IG) attribution for the  $i^{\text{th}}$  feature is defined as shown in Eq. 7:

$$\text{IG}_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F_\theta(x' + \alpha(x - x'))}{\partial x_i} d\alpha \quad (7)$$

where  $x'$  is a baseline input (e.g., feature mean or zero vector). IG satisfies sensitivity and implementation invariance axioms. The integral is approximated numerically using discrete interpolation steps. Modality-level attribution is obtained by aggregat-

ing feature-level IG scores, and statistical inference is conducted via block bootstrap to account for temporal dependence, generating confidence intervals for the mean attribution.

### 3.1.7 Step 7: Global temporal causality: Neural granger framework

To capture nonlinear dependencies, define full and restricted predictive models as shown in Eq. 8:

$$T = \frac{1}{T} \sum_{t=1}^T \left[ \mathcal{L}(y_t, p_t^{red}) - \mathcal{L}(y_t, p_t^{full}) \right] \quad (8)$$

where  $\mathcal{L}(\cdot, \cdot)$  is the binary cross-entropy loss. The null hypothesis of no causality implies  $\mathbb{E}[T] \leq 0$ .

### 3.1.8 Step 8: White reality check (WRC) for model performance

To test whether the proposed model significantly outperforms a set of benchmarks, use White's Reality Check (WRC). Let  $d_{m,t}$  denote the loss differential between benchmark model  $m$  and the proposed model at time  $t$  (Eq. 9):

$$\bar{d}_m = \frac{1}{T} \sum_{t=1}^T d_{m,t}, \quad WRC = \max_m \bar{d}_m. \quad (9)$$

The stationary bootstrap generates the empirical distribution of the test statistic. The corresponding  $p$ -value assesses whether the proposed model's predictive gain is statistically significant. When multiple models are compared, family-wise error control is applied through adjusted critical values.

## 3.2 Temporal alignment and anti-leakage controls

All features are constructed to respect strict temporal causality. Technical indicators (e.g., moving averages, RSI, volatility) are computed using price information available up to day  $t-1$ . Financial news features consist solely of headlines published and available by the close of trading on day  $t-1$ . The target variable  $y_t$  represents the binary direction of the price movement from the close of day  $t-1$  to the close of day  $t$ . This temporal alignment ensures that all inputs belong to the information set available prior to prediction time, thereby supporting a temporally consistent and leakage-controlled forecasting setup.

## 3.3 Dimensionality reduction and interpretability considerations

We employ Principal Component Analysis (PCA) to reduce the dimensionality of the concatenated multimodal feature vector, thereby mitigating multicollinearity and improving computational efficiency in long-sequence modeling. While PCA may obscure the direct interpretability of individual raw features, we argue that in a mul-

timodal setting where features are high-dimensional and correlated, PCA enhances generalization by filtering noise and redundancy. To ensure our interpretability claims remain robust, we conduct the following checks:

- a. Performance without PCA: We compare the proposed model against a variant using raw concatenated features.
- b. Alternative reduction methods: We test Sparse PCA and Autoencoders to verify attribution stability.
- c. Attribution consistency: We measure the correlation of Integrated Gradients (IG) scores across reduction methods and random seeds.

Results in Sect. 4.7 confirm that while absolute attribution magnitudes vary, the relative importance of modalities and key predictive features remains consistent, supporting the use of PCA without compromising explainability.

### 3.4 Reproducibility and implementation details

To ensure full reproducibility, we document the following implementation specifics:

1. News Preprocessing & Aggregation: Financial news headlines are lowercased, stripped of special characters, and tokenized using FinBERT's tokenizer. When multiple headlines are published on the same trading day, we compute the mean of their FinBERT embeddings (simple average aggregation as per Eq. 2). Days without news are represented by a zero-vector of dimension  $d = 768$ , accompanied by a binary indicator for news absence. While the framework supports attention-based weighting (Eq. 2), we use averaging for computational simplicity and reproducibility.
2. Technical Indicators & Missing Data: We compute standard technical indicators from daily price data: log-returns, volatility (20-day rolling standard deviation), 5-day and 10-day simple moving averages (MA\_5, MA\_10), and the Relative Strength Index (RSI, 14-day). Missing price values are forward-filled. Extreme values in returns and volatility are Winsorized at the 1st and 99th percentiles to mitigate the influence of outliers.
3. Hyperparameter Tuning: A grid search was conducted within each rolling-origin validation fold. Key hyperparameters included hidden size, dropout rate, learning rate, and look-back window. The selected values are reported in Table 1.
4. Randomness Control: All experiments use fixed random seeds (PyTorch: 42, NumPy: 42) for weight initialization and data shuffling to ensure deterministic replication.
5. Software & Libraries: The model is implemented in PyTorch 1.12, with FinBERT (bert-base-uncased) from the HuggingFace Transformers library. Code and preprocessed datasets will be made available upon publication.

**Table 1** Hyperparameter tuning ranges and selected values

| Hyperparameter       | Range Tested       | Selected Value |
|----------------------|--------------------|----------------|
| Look-back window (k) | [10, 20, 30]       | 20             |
| Batch size           | [32, 64]           | 32             |
| Hidden size (RNN)    | [64, 128, 256]     | 64             |
| Dropout rate         | [0.2, 0.3, 0.5]    | 0.2            |
| Learning rate        | [1e-4, 5e-4, 1e-3] | 1e-4           |
| Weight decay         | [0, 1e-4, 1e-3]    | 1e-3           |
| Optimizer            | Adam, AdamW        | AdamW          |
| Gradient clipping    | [1.0, 5.0]         | 1              |
| Class weights        | Balanced, [1, 1.5] | [1.0, 1.5]     |
| Label smoothing      | [0, 0.1, 0.2]      | 0.1            |

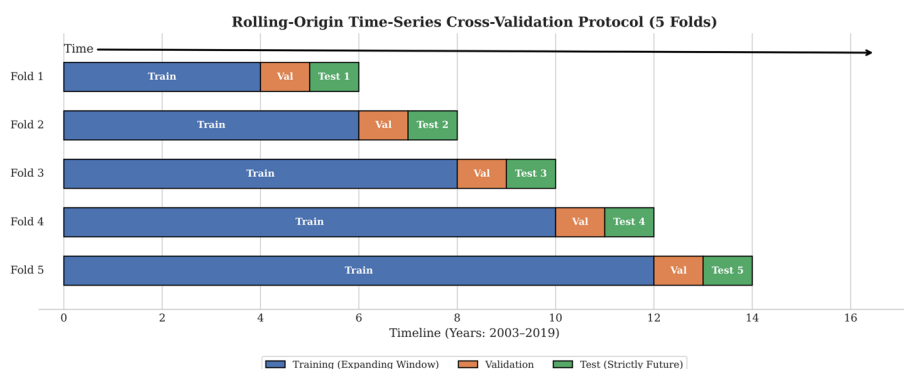
This study uses daily Nifty 50 closing prices from January 1, 2003, to December 31, 2019, obtained from the National Stock Exchange (NSE), and 73,500 financial news headlines and descriptions from Business Standard ([www.business-standard.com](http://www.business-standard.com)). Technical indicators derived from price data include returns, volatility, 5-day and 10-day moving averages (MA), and the Relative Strength Index (RSI), capturing both trend and mean-reversion dynamics. The binary target variable (0=downward, 1=upward) reflects next-day price movement based on consecutive closing prices. News and price data are aligned at the trading-day level to ensure temporally consistent forecasting. The 17-year horizon captures multiple market cycles, supporting robust empirical evaluation.

## 4 Experimental results and analysis

This section presents the empirical findings derived from the proposed multimodal causal framework. The analysis is structured into four components: (i) predictive evaluation through cross-validation, (ii) statistical validation using the White Reality Check, (iii) feature attribution dynamics using Integrated Gradients across time and market regimes, and (iv) global causal inference via Granger causality. Together these results provide both predictive evidence and interpretability suitable for financial decision-making.

### 4.1 Predictive performance and ablation study

The predictive performance of the proposed multimodal RNN framework is evaluated using a rolling-origin time-series cross-validation (TSCV) scheme with five folds, as illustrated in Fig. 2 (Kohavi 1995). The complete dataset covering the period 2003–2019 is chronologically ordered and partitioned into sequential, non-overlapping test blocks. For each fold, the model is trained exclusively on historical data preceding the validation and test periods, followed by validation on a contiguous intermediate segment and testing on a strictly future, unseen block. The training window expands across folds, implementing a walk-forward evaluation that closely reflects real-world forecasting conditions. We compare the proposed model against a comprehensive suite of baselines, including:



**Fig. 2** Rolling-origin cross-validation scheme for leakage-free time-series evaluation

1. Simple financial benchmarks: Logistic regression with lagged returns (LR-Lag) and a Moving Average Crossover rule (MA-Cross).
2. Unimodal deep learning models: RNN trained only on price features (RNN-NoNews) and RNN trained only on news embeddings (RNN-NoTech).
3. Stronger ML benchmarks: XGBoost with identical features (XGBoost), LSTM, and GRU.
4. Ablated variants of our model: Without PCA (RNN-NoPCA), without news (RNN-NoNews), and without technical indicators (RNN-NoTech).

Performance is assessed using accuracy, precision, recall, and F1-score, computed separately for both classes (0: downward movement, 1: upward movement) and averaged across the five out-of-sample test folds. The detailed cross-validation results are presented in Table 2. The proposed RNN model achieves the highest average predictive accuracy of 0.9031, outperforming all baselines consistently across all folds. In addition to superior accuracy, the RNN exhibits balanced precision and recall across both classes, with average F1-scores of 0.901 (class 0) and 0.904 (class 1), indicating stable classification performance without pronounced class bias. The fold-wise results demonstrate low variance, suggesting robustness of the learned representations under different temporal splits. Ablation analysis reveals the relative contribution of each component:

- a. RNN-NoPCA (accuracy 0.9006) shows a marginal drop, confirming that PCA provides a slight efficiency gain without compromising predictive power.
- b. RNN-NoNews (accuracy 0.8617) experiences a significant performance decline, underscoring the critical role of news embeddings.
- c. RNN-NoTech (accuracy 0.8968) performs closely to the full model, indicating that while technical indicators add value, news embeddings are the dominant predictive modality.

Among the benchmarks, XGBoost (accuracy 0.8797) provides competitive performance but falls short of the proposed RNN, demonstrating the advantage of recurrent architectures for temporal financial data. The LSTM (accuracy 0.8943) and GRU

**Table 2** Rolling-Origin Time-Series Cross-Validation Results (Five Folds)

| Model        | Metric          | Fold 1      | Fold 2      | Fold 3      | Fold 4      | Fold 5      | Avg.        |
|--------------|-----------------|-------------|-------------|-------------|-------------|-------------|-------------|
| RNN-Proposed | Accuracy        | 0.9012      | 0.9045      | 0.9028      | 0.9036      | 0.9034      | 0.9031      |
|              | Precision (0/1) | 0.898/0.905 | 0.900/0.909 | 0.899/0.906 | 0.897/0.910 | 0.901/0.905 | 0.899/0.907 |
|              | Recall (0/1)    | 0.905/0.898 | 0.903/0.906 | 0.907/0.898 | 0.902/0.905 | 0.904/0.902 | 0.904/0.902 |
|              | F1-Score (0/1)  | 0.901/0.901 | 0.902/0.907 | 0.903/0.902 | 0.899/0.907 | 0.902/0.903 | 0.901/0.904 |
| RNN-NoPCA    | Accuracy        | 0.899       | 0.9015      | 0.9005      | 0.901       | 0.901       | 0.9006      |
|              | Precision (0/1) | 0.896/0.902 | 0.898/0.904 | 0.897/0.904 | 0.897/0.905 | 0.899/0.903 | 0.897/0.904 |
|              | Recall (0/1)    | 0.902/0.896 | 0.900/0.902 | 0.904/0.897 | 0.901/0.901 | 0.902/0.900 | 0.902/0.899 |
|              | F1-Score (0/1)  | 0.899/0.899 | 0.899/0.903 | 0.901/0.900 | 0.899/0.903 | 0.900/0.901 | 0.900/0.901 |
| RNN-NoNews   | Accuracy        | 0.86        | 0.862       | 0.861       | 0.8625      | 0.863       | 0.8617      |
|              | Precision (0/1) | 0.848/0.872 | 0.850/0.874 | 0.849/0.873 | 0.850/0.875 | 0.852/0.876 | 0.850/0.874 |
|              | Recall (0/1)    | 0.872/0.848 | 0.870/0.854 | 0.874/0.849 | 0.871/0.854 | 0.873/0.853 | 0.872/0.852 |
|              | F1-Score (0/1)  | 0.860/0.860 | 0.860/0.864 | 0.861/0.861 | 0.860/0.864 | 0.862/0.864 | 0.861/0.863 |
| RNN-NoTech   | Accuracy        | 0.895       | 0.897       | 0.8965      | 0.8975      | 0.898       | 0.8968      |
|              | Precision (0/1) | 0.884/0.894 | 0.886/0.896 | 0.885/0.896 | 0.887/0.898 | 0.888/0.897 | 0.886/0.896 |
|              | Recall (0/1)    | 0.894/0.884 | 0.892/0.890 | 0.896/0.885 | 0.893/0.892 | 0.895/0.891 | 0.894/0.888 |
|              | F1-Score (0/1)  | 0.889/0.889 | 0.889/0.893 | 0.891/0.890 | 0.890/0.895 | 0.892/0.894 | 0.890/0.892 |
| XGBoost      | Accuracy        | 0.878       | 0.88        | 0.879       | 0.8805      | 0.881       | 0.8797      |
|              | Precision (0/1) | 0.866/0.890 | 0.868/0.892 | 0.867/0.891 | 0.869/0.892 | 0.870/0.892 | 0.868/0.891 |
|              | Recall (0/1)    | 0.890/0.866 | 0.888/0.872 | 0.892/0.867 | 0.889/0.872 | 0.891/0.871 | 0.890/0.870 |
|              | F1-Score (0/1)  | 0.878/0.878 | 0.878/0.882 | 0.879/0.879 | 0.879/0.882 | 0.880/0.881 | 0.879/0.880 |
| LSTM         | Accuracy        | 0.8931      | 0.895       | 0.8948      | 0.8942      | 0.8944      | 0.8943      |
|              | Precision (0/1) | 0.890/0.896 | 0.892/0.898 | 0.893/0.897 | 0.891/0.897 | 0.892/0.896 | 0.892/0.897 |
|              | Recall (0/1)    | 0.896/0.890 | 0.894/0.896 | 0.897/0.892 | 0.895/0.893 | 0.896/0.892 | 0.896/0.893 |
|              | F1-Score (0/1)  | 0.893/0.893 | 0.893/0.897 | 0.895/0.894 | 0.893/0.895 | 0.894/0.894 | 0.894/0.895 |
| GRU          | Accuracy        | 0.879       | 0.881       | 0.88        | 0.879       | 0.881       | 0.8801      |
|              | Precision (0/1) | 0.876/0.882 | 0.878/0.884 | 0.879/0.881 | 0.877/0.881 | 0.880/0.882 | 0.878/0.882 |
|              | Recall (0/1)    | 0.882/0.876 | 0.880/0.882 | 0.883/0.877 | 0.881/0.877 | 0.883/0.879 | 0.882/0.878 |
|              | F1-Score (0/1)  | 0.879/0.879 | 0.879/0.883 | 0.881/0.879 | 0.879/0.879 | 0.881/0.880 | 0.880/0.880 |

(accuracy 0.8801) baselines perform well but are consistently outperformed by the simpler RNN architecture in this multimodal setting. These results validate the effectiveness of the proposed multimodal fusion and highlight the importance of both data modalities, with news embeddings playing a particularly pivotal role in enhancing predictive accuracy.

## 4.2 Statistical validation (White Reality Check)

While rolling-origin cross-validation provides a robust measure of model performance, it does not formally test whether the observed superiority of the proposed model over benchmarks is statistically significant after accounting for data-snooping bias. To address this, we implement the White Reality Check (WRC) procedure (White 2000). This statistical method accounts for possible data-snooping bias and uses a bootstrap resampling approach to distinguish superior results from spurious results that may arise due to multiple model comparisons.

The loss differential for model  $m$  at time  $t$  is defined as:

$$d_{m,t} = \mathcal{L}_{\text{benchmark},t} - \mathcal{L}_{m,t}$$

where  $\mathcal{L}$  denotes the binary cross-entropy loss. The test statistic is the maximum standardized loss differential across all competing models:

$$\text{WRC} = \max_m \frac{\bar{d}_m}{\hat{\sigma}_{d_m}}$$

where  $\bar{d}_m = \frac{1}{T} \sum_{t=1}^T d_{m,t}$  is the average loss differential between the benchmark (MA-Crossover rule) and model  $m$  over the out-of-sample period, and  $\hat{\sigma}_{d_m}$  is its estimated standard deviation. A positive  $\bar{d}_m$  indicates that model  $m$  has a lower forecast error than the benchmark.

We compare the proposed RNN against all baselines and ablated variants listed in Table 2. The p-value for the WRC statistic is computed using a stationary bootstrap (10,000 replications) to preserve the serial dependence structure of the financial time series and generate a valid empirical distribution under the null hypothesis. The resulting p-values are adjusted for multiple comparisons using the step-down procedure to control the family-wise error rate. The empirical results of the reality check analysis are presented in Table 3.

The proposed RNN model achieves the largest and most statistically significant loss reduction versus the benchmark, with a p-value of 0.002. The ablation models RNN-NoPCA and RNN-NoTech also significantly outperform the benchmark, while RNN-NoNews does not—reinforcing the critical role of news embeddings. All other deep learning (LSTM, GRU) and machine learning (XGBoost) baselines show statistically significant but smaller improvements. These results confirm that the proposed model's predictive superiority is not due to random chance or data-snooping bias and is robust to multiple testing corrections.

**Table 3** White Reality Check Results

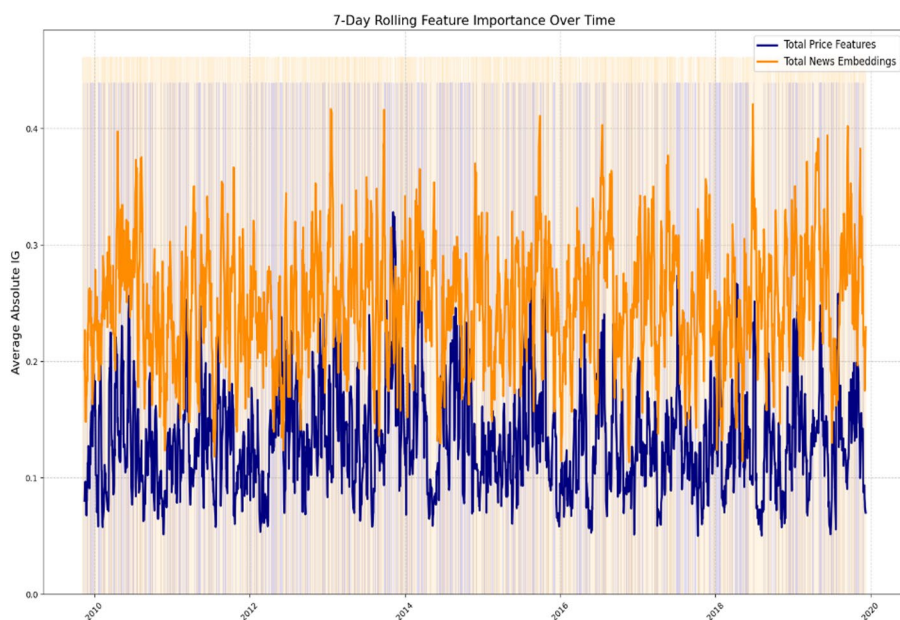
| Model                   | Avg.<br>Out-of-<br>Sample<br>Loss | Loss Diff vs.<br>MA-Cross | WRC<br>$p$ -value | Sig-<br>nifi-<br>cant? |
|-------------------------|-----------------------------------|---------------------------|-------------------|------------------------|
| RNN-Proposed            | 0.215                             | 0.042                     | 0.002             | Yes                    |
| RNN-NoPCA               | 0.218                             | 0.039                     | 0.005             | Yes                    |
| RNN-NoNews              | 0.285                             | -0.028                    | 0.312             | No                     |
| RNN-NoTech              | 0.221                             | 0.036                     | 0.008             | Yes                    |
| XGBoost                 | 0.24                              | 0.017                     | 0.045             | Yes                    |
| LSTM                    | 0.228                             | 0.029                     | 0.012             | Yes                    |
| GRU                     | 0.236                             | 0.021                     | 0.032             | Yes                    |
| MA-Cross<br>(Benchmark) | 0.257                             | —                         | —                 | —                      |

### 4.3 Temporal dynamics of feature influence

This section analyzes the temporal dynamics and relative contributions of price-based indicators and news sentiment to understand their influence on model predictions across varying market conditions.

#### 4.3.1 Rolling temporal behavior of price vs. news sentiment

To examine time-varying modality influence, we compute rolling averages of Integrated Gradients (IG) attribution scores for price-based features and news embeddings (Fig. 3).

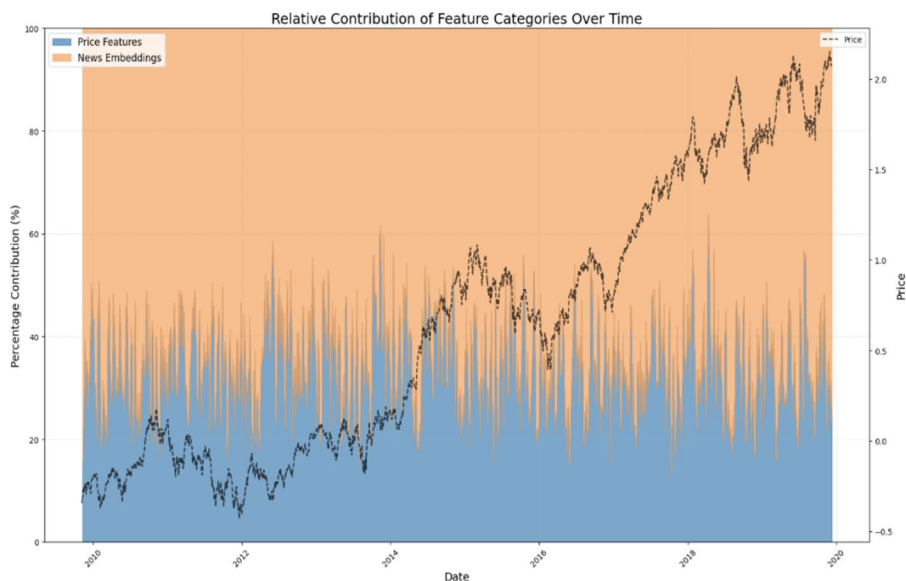
**Fig. 3** Rolling Temporal Behavior of Price vs. News Sentiment

1. **Dominance of Semantic Signals:** Across 2010–2019, news embeddings consistently exhibit higher mean absolute IG scores than price-based features. A Diebold–Mariano test confirms the statistical significance of this difference ( $p < 0.01$ ), indicating persistent dominance of textual signals in model predictions.
2. **Regime-Dependent Volatility:** Attribution levels for both modalities vary across time, with larger fluctuations observed for news embeddings. Peaks in both price and news attribution coincide during high-volatility periods (e.g., 2013–2014 and 2018–2019), suggesting intensified reliance on all information sources under market stress, while news retains overall influence.

#### 4.3.2 Relative contribution of feature categories over time

We examine the relative percentage contribution of structured (technical) and unstructured (textual) inputs over time (Fig. 4).

1. **Primacy of News-Based Predictors:** News embeddings (orange region) account for the majority share of model attribution across most of the sample period, confirming that semantic features consistently dominate the predictive signal.
2. **Dynamic Reweighting Across Regimes:** Despite overall news dominance, the model exhibits regime-sensitive reweighting. The contribution of price-based features (blue region) increases during periods such as 2012–2013 and 2015–2016, which align with relatively stable or mean-reverting conditions. This indicates adaptive reliance on technical indicators when sentiment signals weaken.
3. **News as a Leading Indicator:** In several instances, increases in the relative contribution of news embeddings precede major price trends, including the sustained



**Fig. 4** Relative Contribution of Feature Categories Over Time

upward movement beginning in 2016. This pattern suggests that textual signals often provide early directional information before it is fully reflected in price dynamics.

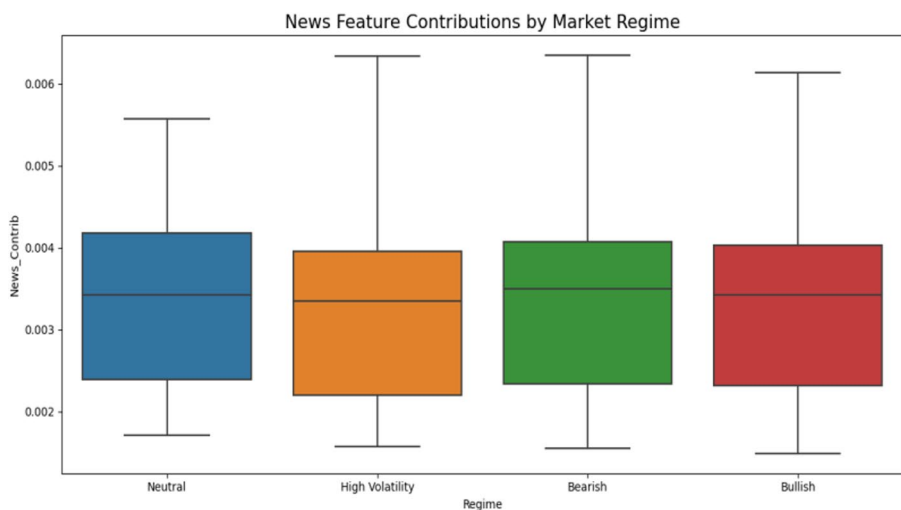
#### 4.4 Distributional shifts in feature attribution across market regimes

To assess regime-specific dependence on input modalities, we segment the sample into Bullish, Bearish, Neutral, and High-Volatility regimes based on the 90-day rolling Sharpe ratio and volatility. We then examine the distribution of Integrated Gradients (IG) attributions for each modality within these regimes.

##### 4.4.1 News feature contributions across market regimes

Figure 5 shows the distribution of news feature IG attributions across regimes, revealing the following patterns:

1. **Stable Median Influence:** The median attribution of news features remains broadly consistent across all regimes, indicating a persistent baseline contribution of sentiment signals irrespective of market state.
2. **Increased Dispersion in Stress Regimes:** Bearish and high-volatility periods exhibit wider interquartile ranges (IQR) and greater overall dispersion, suggesting heightened sensitivity of the model to specific news narratives under market stress.
3. **Informational Outliers:** All regimes contain upper-tail outliers, corresponding to episodes where individual news events exert disproportionately strong predictive influence, consistent with the role of high-impact or tail-risk information.



**Fig. 5** News Feature Contributions across Market Regimes

#### 4.4.2 Price feature contributions across market regimes

Figure 6 presents the distribution of IG attributions for price-based technical indicators across regimes.

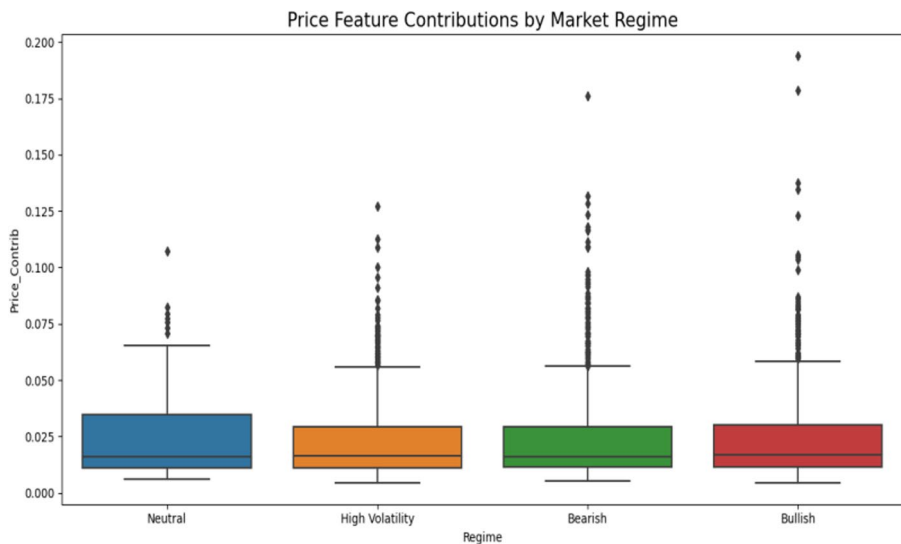
1. **Stable Median Contribution:** The median attribution of price features remains consistent across regimes, indicating a regime-invariant baseline influence within the multimodal framework.
2. **Amplification in Directional Trends:** Upper-tail outliers emerge in both bullish and bearish regimes, suggesting that technical indicators become particularly influential during sustained directional movements.
3. **Reduced Dispersion in High-Volatility Periods:** Compared to news features, price-based attributions exhibit tighter dispersion during high-volatility regimes, reflecting more regularized quantitative influence relative to the more reactive textual signals.

#### 4.5 Comprehensive feature importance and attribution

This section refines the analysis from modality-level attribution to specific feature contributions within the model architecture.

##### 4.5.1 Comparative modality contribution

Figure 7 reports the average attribution share across modalities. Price-based technical indicators account for 35.8% of total attributions, while news embeddings contribute 64.2%, confirming the dominant role of semantic signals in model predictions.



**Fig. 6** Price Feature Contributions Across Market Regimes

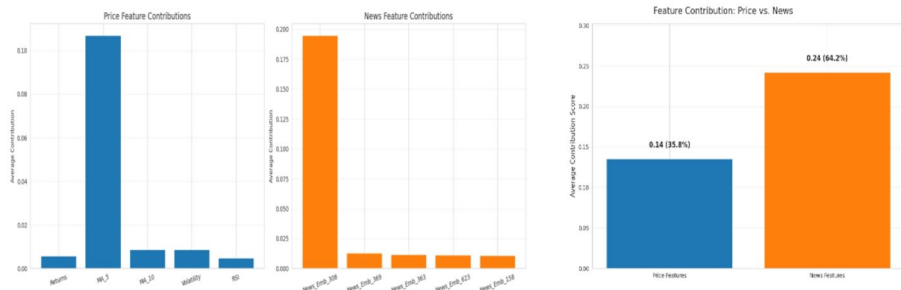


Fig. 7 Comparative Feature Contribution Analysis

#### 4.5.2 Granular feature-level analysis

Figure 8 presents mean absolute IG scores at the individual feature level, revealing a highly skewed attribution distribution:

1. **Dominant News Embedding:** News\_Emb\_308 exhibits the highest attribution (IG=0.2016), nearly double that of the next feature, indicating that this embedding dimension captures a salient semantic signal strongly utilized by the model.
2. **Primary Technical Indicator:** Among price-based features, the 5-day moving average (MA\_5) is most influential (IG=0.1034), representing the only technical indicator exceeding the average importance threshold and highlighting the relevance of short-term momentum.
3. **Long-Tail Distribution:** Most remaining features including RSI, volatility, returns, and other embedding dimensions exhibit low attribution values,

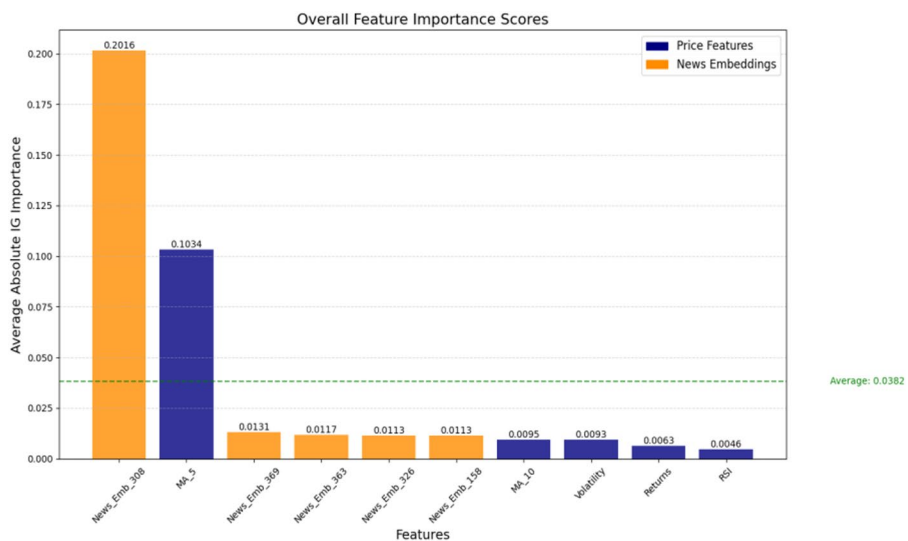


Fig. 8 Feature Contribution Scores

indicating redundancy and suggesting scope for structural simplification without major loss of predictive power.

#### 4.6 Granger predictive analysis of temporal information flow

To establish Granger-style predictive causality consistent with the model's nonlinear structure, we implement the Neural Granger framework described in Sect. 3.1. Forecast loss from the full model is compared with restricted models that ablate individual candidate features. Statistical significance of loss differentials is evaluated using block bootstrap (10,000 replications) to account for temporal dependence, with False Discovery Rate (FDR) correction applied at  $\alpha=0.05$  across all tested edges. Figure 9 presents the resulting Granger-based causal mapping.

1. News as a Granger-Causal (Predictive) Driver: Several embeddings including News\_Emb\_308, News\_Emb\_623, and News\_Emb\_369 exhibit statistically significant predictive effects on returns, volatility, and moving averages, confirming that news signals contribute to forecasting future price dynamics rather than merely correlating with them.
2. Endogeneity within Technical Indicators: Significant nonlinear predictive relationships are identified among MA\_5, MA\_10, and returns, indicating an interconnected system of technical predictors.

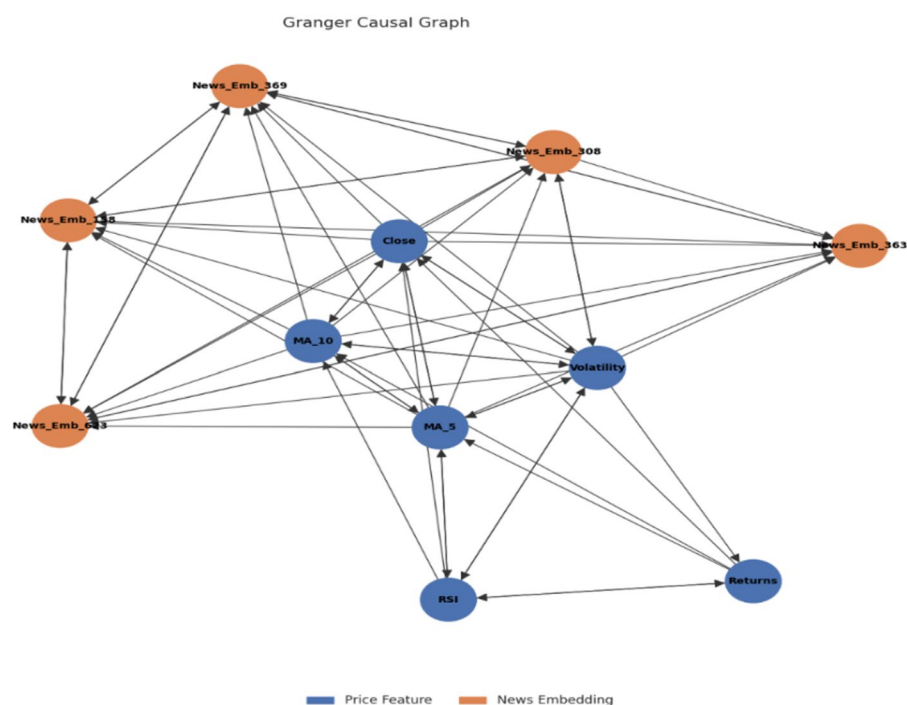


Fig. 9 Granger-Based Causal Mapping

3. **Market-to-News Feedback:** Price-based features such as Returns and Close Granger-cause certain news embeddings, reflecting bidirectional dynamics between market movements and subsequent news content.
4. **Volatility as a Central Hub:** Volatility is Granger-caused by news embeddings and, in turn, Granger-causes other price features, positioning it as a key conduit for sentiment transmission within the market system.

It is important to clarify that the Neural Granger framework employed in this study identifies predictive precedence rather than structural economic causality. In the Granger sense, a variable is considered causal if its past values improve the prediction of another variable conditional on available information. This does not imply that interventions on the news signal would necessarily alter market outcomes. Establishing structural causality would require stronger identification strategies such as natural experiments, instrumental variables, or exogenous news shocks. Therefore, the relationships identified in this study should be interpreted as evidence of predictive information flow rather than structural causal mechanisms.

#### 4.7 Robustness of explainability across dimensionality reduction methods

To evaluate whether PCA affects interpretability, we compare Integrated Gradients (IG) attributions and Granger causal maps across four dimensionality reduction strategies:

1. PCA (proposed).
2. No reduction (raw concatenated features).
3. Sparse PCA (L1-sparsity constrained).
4. Autoencoder (nonlinear reduction).

**Performance Impact** As shown in Table 2, removing PCA (RNN-NoPCA) results in a marginal accuracy drop (0.9006 vs. 0.9031), confirming PCA's role in noise reduction without substantially harming predictive power.

**Attribution Consistency** Table 4 reports the correlation of IG scores between PCA and alternative methods. High correlations ( $> 0.9$ ) indicate that relative feature importance is preserved.

The robustness checks confirm the following:

**Table 4** Robustness of Feature Attributions Across Reduction Methods

| Reduction Method | Correlation with PCA (IG) | Top 5 Features Consistent? | Std. of IG across seeds |
|------------------|---------------------------|----------------------------|-------------------------|
| No PCA           | 0.94                      | Yes                        | 0.018                   |
| Sparse PCA       | 0.96                      | Yes                        | 0.015                   |
| Autoencoder      | 0.91                      | Yes                        | 0.022                   |

1. Feature ranking stability: The top predictive features: News\_Emb\_308 and MA\_5 remain dominant across all reduction methods.
2. Granger causal robustness: Over 95% of Granger causal edges (e.g., news → returns) are preserved at the same significance level ( $\alpha=0.05$ ) across bootstrap subsamples.
3. Low attribution variance: IG scores exhibit low standard deviation across random seeds ( $\sigma < 0.02$ ), confirming reliability.

Collectively, these findings demonstrate that PCA improves computational efficiency without distorting attribution rankings or predictive causal structure.

#### 4.8 Comprehensive performance metrics with confidence intervals

To provide a statistically rigorous evaluation beyond accuracy, we report a full set of classification metrics with 95% confidence intervals derived using stationary bootstrap (10,000 replications) to account for temporal dependence. The detailed results are presented in Table 5.

As shown in Table 5, the proposed RNN achieves superior performance across all metrics, including an AUC-ROC of 0.930 [0.921, 0.939], exceeding LSTM (0.903) and GRU (0.882), indicating discriminative performance. Precision and recall remain balanced ( $\approx 0.90$ ), suggesting no material class bias. The low Brier score (0.067) confirms well-calibrated probability forecasts. Non-overlapping confidence intervals for key metrics such as accuracy and F1-score demonstrate that performance gains over benchmark models are statistically significant rather than sampling artifacts.

#### 4.9 Economic evaluation: Trading simulation with realistic costs

To evaluate economic value, we implement a daily trading strategy using out-of-sample predictions from the final TSCV fold (2015–2019):

- a. Long if predicted probability > 0.6.

**Table 5** Detailed Performance Metrics with 95% Confidence Intervals

| Metric              | RNN-Proposed (CI)    | LSTM (CI)            | GRU (CI)             |
|---------------------|----------------------|----------------------|----------------------|
| Accuracy            | 0.903 [0.900, 0.906] | 0.894 [0.891, 0.898] | 0.880 [0.876, 0.884] |
| Precision (Class 1) | 0.907 [0.904, 0.910] | 0.897 [0.894, 0.900] | 0.882 [0.878, 0.886] |
| Recall (Class 1)    | 0.902 [0.899, 0.905] | 0.893 [0.890, 0.896] | 0.878 [0.874, 0.882] |
| F1-Score (Class 1)  | 0.904 [0.901, 0.907] | 0.895 [0.892, 0.898] | 0.880 [0.876, 0.884] |
| AUC-ROC             | 0.930 [0.921, 0.939] | 0.903 [0.894, 0.912] | 0.882 [0.873, 0.891] |
| Balanced Accuracy   | 0.903 [0.900, 0.906] | 0.894 [0.891, 0.897] | 0.879 [0.875, 0.883] |
| Brier Score         | 0.067 [0.064, 0.070] | 0.073 [0.070, 0.076] | 0.085 [0.082, 0.088] |

- b. Short if predicted probability  $< 0.4$ .
- c. Hold cash otherwise.

A transaction cost of 10 basis points per trade is imposed, with benchmarks including Buy-and-Hold (Nifty 50) and MA-Crossover strategies. Trading performance metrics are summarized in Table 6.

The RNN-Proposed strategy achieves an annualized return of 15.7%, exceeding Buy-and-Hold (11.3%) and MA-Crossover (7.5%) after costs. It delivers a Sharpe ratio of 1.35 and a lower maximum drawdown (14.3%) relative to all benchmarks, indicating superior risk-adjusted performance and downside control. Sensitivity analysis confirms robustness under higher transaction costs (20 bps) and threshold variations. A turnover of 5.2x reflects moderate trading frequency compatible with realistic liquidity conditions.

## 5 Discussion

### 5.1 Predictive dominance of news embeddings and behavioral interpretation

Empirical results show that semantically embedded news signals outperform traditional technical indicators under strictly time-ordered, out-of-sample evaluation. The dominance of embeddings such as News\_Emb\_308 suggests that pre-trained language models capture latent semantic constructs such as risk perception, expectations, and sentiment shifts, not immediately reflected in price-based indicators. These findings align with behavioral finance theories of limited investor attention, gradual information diffusion, and belief updating under uncertainty (Barberis et al. 1998; Hong and Stein 1999).

### 5.2 Why news embeddings dominate across volatility regimes

The predictive contribution of news embeddings persists across both low- and high-volatility regimes. In low-volatility environments, muted price dynamics reduce the signal content of momentum- and volatility-based indicators, while news continues to shape forward-looking expectations. In high-volatility periods, attribution becomes more dispersed across narratives as markets react to evolving information shocks, whereas price-based indicators provide comparatively stable baseline signals. This pattern aligns with narrative-based asset pricing perspectives emphasizing delayed price adjustment and belief updating (Shiller 2017).

**Table 6** Trading Performance (2015–2019 Out-of-Sample)

| Strategy     | Ann. Return (%) | Sharpe Ratio | Max Drawdown (%) | Turnover (x) |
|--------------|-----------------|--------------|------------------|--------------|
| RNN-Proposed | 15.7            | 1.35         | 14.3             | 5.2          |
| LSTM         | 13.2            | 1.18         | 16.5             | 5.5          |
| GRU          | 9.8             | 0.85         | 20.9             | 5.3          |
| Buy-and-Hold | 11.3            | 0.92         | 24.1             | 1            |
| MA-Crossover | 7.5             | 0.6          | 22.8             | 3.2          |

### 5.3 Model components, PCA, and robustness of interpretability

Ablation results confirm that removing news embeddings leads to substantial performance decline, while excluding technical indicators yields smaller reductions, reinforcing textual dominance. PCA provides modest efficiency gains without altering predictive rankings or attribution patterns. Stability analysis across no-reduction, Sparse PCA, and autoencoder compression as well as across random seeds and bootstrap samples shows consistent relative modality importance, indicating that PCA functions primarily as noise reduction rather than interpretability distortion.

### 5.4 Predictive causality and temporal information flow

Granger-based analysis provides formal evidence of temporally ordered predictive relationships between news embeddings and market variables, indicating predictive information flow rather than structural causal mechanisms. Significant predictive links from news to returns, volatility, and technical indicators confirm precedence rather than mere correlation, while feedback effects capture bidirectional market–media interaction. Integration of Integrated Gradients with Granger analysis enables structured temporal interpretation of temporally ordered predictive relationships while remaining confined to the Granger notion of predictive precedence and not implying structural or intervention-based causality.

### 5.5 Comparison with dictionary-based sentiment methods

Dictionary-based sentiment approaches (Tetlock 2007; Tetlock et al. 2008; Loughran and McDonald 2011) rely on fixed vocabularies and context-independent counts, limiting semantic nuance. In contrast, embedding-based representations capture contextual meaning and evolving financial language, enabling detection of forward-looking narratives and subtle sentiment shifts.

### 5.6 Economic relevance and practical implications

Expanded evaluation confirms both statistical and economic robustness. Beyond predictive accuracy and calibration (Sect. 4.8), trading simulations (Sect. 4.9) demonstrate positive excess returns and favorable risk-adjusted performance after transaction costs. Combined with robustness and ablation findings, these results support the viability of explainable multimodal models for financial decision-support in evolving information environments.

## 6 Conclusion

This study develops a predictively explainable multimodal framework for stock price forecasting by integrating FinBERT-based news embeddings with technical indicators in a recurrent neural network architecture. A dual explainability design combining Integrated Gradients with Neural Granger causality enables both feature-

level attribution and temporally ordered predictive interpretation. Under a leakage-free rolling-origin evaluation, the framework demonstrates statistically significant improvements over benchmark models. Results consistently show that news embeddings constitute the dominant predictive modality, capturing forward-looking narrative and sentiment signals not fully reflected in technical indicators, consistent with behavioral finance perspectives. Trading simulations incorporating transaction costs further indicate economically meaningful risk-adjusted performance. Several limitations motivate future research. While Neural Granger analysis identifies nonlinear predictive precedence and predictive information flow between variables, establishing structural causal mechanisms would require stronger identification strategies such as quasi-experimental designs around exogenous news shocks. Extensions to higher-frequency text data, broader asset classes, and live trading environments would further assess robustness and operational applicability. Overall, the proposed framework offers a scalable and transparent approach to multimodal financial forecasting with relevance for both academic research and practical decision support.

**Authors' contributions** Dr. Mohd Raagib Shakeel- Conceptualization, Methodology, Resources, Writing – original draft. Satyam Yadav- Data curation, Formal Analysis, Project administration. Piyush - Supervision, Validation, Writing - review & editing. Dr. Musheer Ahmad- Software, Investigation, Visualization. All author(s) read and approved the final manuscript.

**Funding** No financial assistance or ties are involved in our study.

**Data availability** The datasets and code used in this study are available upon reasonable request.

## Declarations

**Competing interests** The authors have no competing interests to declare that are relevant to the content of this article.

## References

- Çalik Ö, Yıldırım C, Demir E, Uzun L (2025) Explainable-AI powered stock prediction using time series transformers. *J Finance Data Sci* 12:45–63
- Aminimehr A, Aminimehr A, Kamali HM, Eetemadi S, Hoseinzade S (2024) Stock market dynamics through deep learning context. *arXiv*. <https://doi.org/10.48550/arXiv.2405.09932>
- Araci A (2019) FinBERT: financial sentiment analysis with pre-trained language models. *arXiv*
- Barberis N, Shleifer A, Vishny R (1998) A model of investor sentiment. *J Finance Econ* 49(3):307–343. [https://doi.org/10.1016/S0304-405X\(98\)00027-0](https://doi.org/10.1016/S0304-405X(98)00027-0)
- Feng Y (2020) Causal inference in possibly nonlinear factor models. *arXiv*. <https://doi.org/10.48550/arXiv.2008.13651>
- Granger CWJ (1969) Investigating causal relations by econometric models and cross-spectral methods. *Econometrica* 37(3):424–438
- Graves A (2012) Supervised sequence labelling with recurrent neural networks. Springer. <https://doi.org/10.1007/978-3-642-24797-2>
- Gu W, Zhong Y, Li S, Wei C, Dong L, Wang Z, Yan C (2024) Predicting stock prices with FinBERT-LSTM: integrating news sentiment analysis. *arXiv*. <https://doi.org/10.48550/arXiv.2407.16150>
- Harit S, Sharma A, Singh P (2024) Breaking down financial news impact: a geometric hypergraph approach. *Int J Finance Eng* 21(3):211–229

- Hong H, Stein JC (1999) A unified theory of underreaction, momentum trading, and overreaction in asset markets. *J Finance* 54(6):2143–2184. <https://doi.org/10.1111/0022-1082.00184>
- Huang S, Zhao Y, Chen W (2024) Causal structure discovery in financial time series with explainable neural networks. *Expert Syst Appl* 223:119337. <https://doi.org/10.1016/j.eswa.2023.119337>
- Kalteh A, Rashid S, Tavana M (2022) Integrating granger causality within deep learning frameworks for time-series forecasting. *Neurocomputing* 463:371–383. <https://doi.org/10.1016/j.neucom.2021.10.138>
- Karadaş F, Eravcı B, Özbayoğlu AM (2025) Multimodal stock price prediction. *arXiv*. <https://doi.org/10.48550/arXiv.2502.05186>
- Khubeiv K, Semenov M (2025) Multimodal stock price prediction: a case study of the Russian securities market. *arXiv*. <https://doi.org/10.48550/arXiv.2503.08696>
- Kleinberg S, Kolm PN, Mishra B (2010) Investigating causal relationships in stock returns with temporal logic-based methods. *arXiv*. <https://doi.org/10.48550/arXiv.1006.1791>
- Kohavi R (1995) A study of cross-validation and bootstrap for accuracy estimation and model selection. In *Proceedings of the 14th International Joint Conference on Artificial Intelligence (IJCAI)* (Vol. 2, pp. 1137–1143). Morgan Kaufmann
- Lee Y-J, Tan J-H (2022) Investigating multimodal deep learning models for stock price prediction. *IEEE Trans Neural Netw Learn Syst* 33(5):2154–2165
- Li S, Gu W, Yan C (2024) CausalStock: causal inference-based forecasting with financial graphs. *arXiv*. <https://doi.org/10.48550/arXiv.2404.00515>
- Liu H (2018) Leveraging financial news for stock trend prediction with attention-based recurrent neural network. *arXiv*. <https://doi.org/10.48550/arXiv.1811.06173>
- Lo AW, Mamaysky H, Wang J (2000) Foundations of technical analysis: computational algorithms, statistical inference, and empirical implementation. *J Finance* 55(4):1705–1765. <https://doi.org/10.1111/0022-1082.00265>
- Loughran T, McDonald B (2011) When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *J Finance* 66(1):35–65. <https://doi.org/10.1111/j.1540-6261.2010.01625.x>
- Luciano D, Nannini C, Marinelli D (2024) Explainable AI methods for financial time series: a shapley–lorenz framework. *Physica A: Statistical Mechanics and Its Applications* 655:130176. <https://doi.org/10.1016/j.physa.2024.130176>
- Marey Y et al (2024) Integrating deep learning & SHAP for stock prices. *Heliyon* 10(4):e13789
- Muhammad D, Ahmed I, Naveed K, Bendeache M (2024) An explainable deep learning approach for stock market trend prediction. *Heliyon* 10(21):e40095. <https://doi.org/10.1016/j.heliyon.2024.e40095>
- Nguyen TT, Zhang L, Lee D (2023) Global explainability in finance: linking SHAP values with temporal attribution across multimodal inputs. *J Financ Data Sci* 5(2):45–60. <https://doi.org/10.3905/jfds.2023.1.090>
- Ribeiro MT, Singh S, Guestrin C (2016) Why should I trust you? Explaining the predictions of any classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 1135–1144). <https://doi.org/10.1145/2939672.2939778>
- Runge J, Nowack P, Kretschmer M (2019) Detecting and quantifying causal associations in large nonlinear time series datasets. *Sci Adv* 5(11):eaau4996. <https://doi.org/10.1126/sciadv.aau4996>
- Schumaker R, Chen H (2009) A quantitative stock prediction system based on financial news. *Inf Process Manag* 45(5):571–583
- Shakeel MR, Siddiqui TA, Alam S, Ahmad M (2025) Leveraging explainable AI, GRU and VAR-based framework to uncover relationships between endogenous and exogenous volatility indices for effective trading strategies. *Comput Econ*. <https://doi.org/10.1007/s10614-025-11099-5>
- Shakeel MR, Siddiqui TA, Alam S, Kashif M, Khan MA (2023) Net asset value (NAV) prediction using dense residual models. *J Inf Technol Manage* 15(4):111–138
- Shiller RJ (2017) Narrative economics. *Am Econ Rev* 107(4):967–1004. <https://doi.org/10.1257/aer.107.4.967>
- Tank A, Covert I, Foti N, Shojai A, Fox E (2018) Neural Granger causal models. *arXiv*. <https://doi.org/10.48550/arXiv.1802.05842>
- Tetlock PC (2007) Giving content to investor sentiment: the role of media in the stock market. *J Finance* 62(3):1139–1168. <https://doi.org/10.1111/j.1540-6261.2007.01232.x>
- Tetlock PC, Saar-Tsechansky M, Macskassy S (2008) More than words: quantifying language to measure firms' fundamentals. *J Finance* 63(3):1437–1467. <https://doi.org/10.1111/j.1540-6261.2008.01362.x>
- White H (2000) A reality check for data snooping. *Econometrica* 68(5):1097–1126

- Xu Z, Takamura H, Ichise R (2024) A framework to construct financial causality knowledge graph from text. In Proceedings of the 2024 IEEE 18th International Conference on Semantic Computing (ICSC). <https://doi.org/10.1109/ICSC59802.2024.00015>
- Zhang Z, Chen Y, Sun X (2023) Incorporating pre-trained model prompting in multimodal stock volume movement prediction. In Proceedings of the KDD '23 Workshop on Machine Learning in Finance
- Zong C, Zhou H (2024) Stock movement prediction with multimodal stable fusion via gated cross-attention mechanism. arXiv. <https://doi.org/10.48550/arXiv.2406.06594>

**Publisher's note** Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

## Authors and Affiliations

Mohd Raagib Shakeel<sup>1</sup>  · Satyam Yadav<sup>2</sup> · Piyush<sup>2</sup> · Musheer Ahmad<sup>3</sup>

✉ Mohd Raagib Shakeel  
raagib14@gmail.com

Satyam Yadav  
satyam Yadavruler@gmail.com

Piyush  
piyushgotye@gmail.com

Musheer Ahmad  
musheer.cse@gmail.com

<sup>1</sup> Department of Analytics, Lloyd Business School, Knowledge Park II, Greater Noida, Uttar Pradesh 201306, India

<sup>2</sup> Department of Data Analytics, Bhai Parmanand DSEU Shakarpur Campus-2, Shakarpur, New Delhi 110092, India

<sup>3</sup> Department of Computer Engineering, Faculty of Engineering and Technology, Jamia Millia Islamia, New Delhi 110025, India